

Power data

Windspeed	Blade angle	Rotorspeed	Power output
5	11	125	48
7	9	155	72
9	15	175	101
11	13	220	142
13	18	235	176

$$X = \begin{bmatrix} 1 & 5 & 11 & 125 \\ 1 & 7 & 9 & 155 \\ 1 & 9 & 15 & 175 \\ 1 & 11 & 13 & 220 \\ 1 & 13 & 18 & 235 \end{bmatrix}$$

$$Y = \begin{bmatrix} 48 \\ 72 \\ 101 \\ 142 \\ 176 \end{bmatrix}$$

$$\theta = (X^T X)^{-1} X^T Y$$

$$X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 5 & 7 & 9 & 11 & 13 \\ 11 & 9 & 15 & 13 & 18 \\ 125 & 155 & 175 & 220 & 235 \end{bmatrix} \begin{bmatrix} 1 & 5 & 11 & 125 \\ 1 & 7 & 9 & 155 \\ 1 & 9 & 15 & 175 \\ 1 & 11 & 13 & 220 \\ 1 & 13 & 18 & 235 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 45 & 66 & 910 \\ 45 & 445 & 630 & 8760 \\ 66 & 630 & 920 & 12485 \\ 910 & 8760 & 12485 & 173900 \end{bmatrix}$$

$$[X^T X]^{-1} = \begin{bmatrix} 76.9 & 18.21 & -8.1 & -1.10 \\ 18.21 & 4.58 & -0.72 & -0.27 \\ -8.1 & -0.72 & 0.16 & 0.04 \\ -1.10 & -0.27 & 0.04 & 0.01 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 539 \\ 5503 \\ 7703 \\ 107435 \end{bmatrix}$$

$$(X^T X) : (X^T y) =$$

99.9112

11.2174

2.4166

0.9058